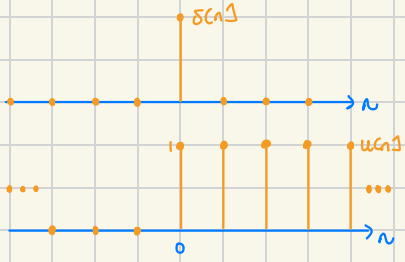
ELG 3125 - Signals & Systems

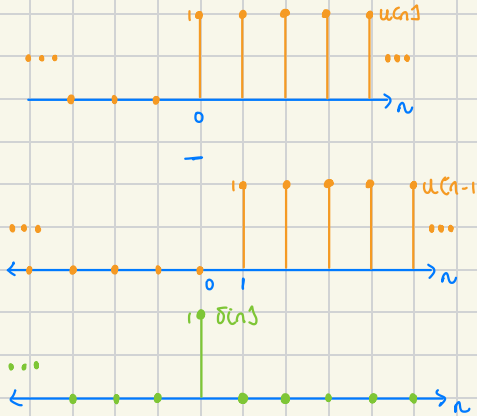
Lecture 3 - Part 1

Discrete-time unit impulse: $\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$



Discrete-time unit step: $u[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$

$$\delta[n] = u[n] - u[n-1]$$

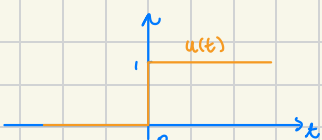


$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

Important property: $x[n]\delta[n] = x[0]\delta[n]$

$$x[n]\delta[n-n_0] = x[n_0]\delta[n-n_0]$$

Continuous-time unit step: $u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$



$$\delta(t) = \frac{du(t)}{dt} \rightarrow du(t) = \delta(t)\,dt$$

$$\rightarrow u(t) = \int du(t) = \int \delta(t)\,dt$$

Sampling property of the continuous-time unit impulse

$$x(t)\delta(t) = x(0)\delta(t)$$

or: $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$

Continuous-time systems and discrete-time systems

